

JTAC Protocols - 24th STS

Basic Requirements

-High level of map reading skills.

This includes grid references.

-Radio comms.

This seems redundant, but the ability to provide quality information with brevity is paramount to these roles. Don't use three words if you can use one.

-Patience and Maturity.

You might get caught in a shit sandwich between command and another asset. You might not have anything to do because there's nothing going on, or the commander doesn't want outside assets to kill anything else. Patience and maturity is required to succeed and be invited back to the role.

-An Arma JTAC is a curator, sometimes more so than the mission maker themselves.

The JTAC has to curate an experience for air AND ground. Just because you see it, and CAN have assets kill it, doesn't mean you should. Can infantry kill it? Do we have armour that needs some action? Artillery battery doing f*ck all back at base? Refer to the line labelled "Patience and Maturity".

-Pen and Paper.

Lots of information will be passed back and forth. Even the best of us can't keep track of it all in our heads, so have a notebook handy and make use of it.

Advanced Comms & Brevity

Both RTOs and JTACs can benefit from brevity and use of advanced comms procedures.

"Hey you, this is me", "OVER", "OUT", "WAIT ONE" should be standard practice at this point.

Clarity of speech, pronunciation, pace and rhythm contribute to good comms. Saying all the right things too fast, or mumbled, helps no one.

Warno's are short transmissions used to get the other callsigns attention, and make them aware of what you want to actually say, but allows them to get ready first. "Interrogative", "Message", or nothing at all are some examples.

Vanguard: "Buckshot, Vanguard, SITREP, OVER"

Buckshot: "Vanguard, Buckshot, interrogative, OVER"

"BREAK" is used during longer transmissions, or as a pause. The net is not cleared for other traffic, as there is more to follow from the current callsign. Very useful if relaying information, or addressing multiple callsigns.

Vanguard: "Vanguard, copy, A-10 Charlie with 4x Maverick, 12x Paveway, 19x Hydra, and

guns, time on station 40 minutes. **BREAK** *pause* “bla bla bla provides tasking or situational update”

CAS Controller

Being the designated CAS Controller requires more multitasking and comms abilities than probably any other role in the game. Even those of us who are really good at it, step on our own tongues and make T-Shirts quoting “Rodin Omeo”. It’s incredibly rewarding when you get it right, and table tossing when you don’t. My personal recommendation is to start seeing the pieces on the chessboard in game. Ignore individual players, and move up a step to squads, elements, and other assets. Start to see the moves they’re making, and *might* be making, and then start to anticipate how CAS might play into that picture.

Glossary of Terms

*JTAC/Controller	-Ground personnel who provides final say on CAS delivery
*IP	-Ingress/Initial Point (starting vector for CAS runs)
*BP	-Battle Point (hover point for rotary to engage from)
*HA	-Holding Area (area of relative safety for rotary to orbit)
CP	-Control Point (fixed wing hold point/reference point)
*Egress	-Direction or method to leave after ordinance drop
*BDA	-Battle Damage Assessment (target destroyed? % of destroyed?)
Paveway	-Laser guided bomb (most commonly GBU-12 500lbs)
Maverick	-AGM 65 Air-To-Ground missile
Hellfire	-AGM 114 Air-To-Ground missile
Heater	-AIM-9 Sidewinder Air-To-Air missile
HEDP	-High Explosive Dual Purpose (big fire, good for infantry)
HEAT	-High Explosive Anti Tank (smaller fire, penetration for armour)
WP	-Willy Pete (white phosphorus, very good for infantry and smoke cover)
**“Pickle”	-Bomb weapons release by air crew
**“Rifle”	-AGM weapons release by air crew
*“Guns”	-Guns/cannon fired by air crew
*“RTB”	-Return to Base
“Bingo”	-Out of fuel/minimum fuel for RTB
“Winchester”	-Out of ordinance

*indicates term can be expected to be in common use

**indicates term is mandatory for usage

Weapons Types

Far more munitions types exist, but these are the most common to encounter and use.

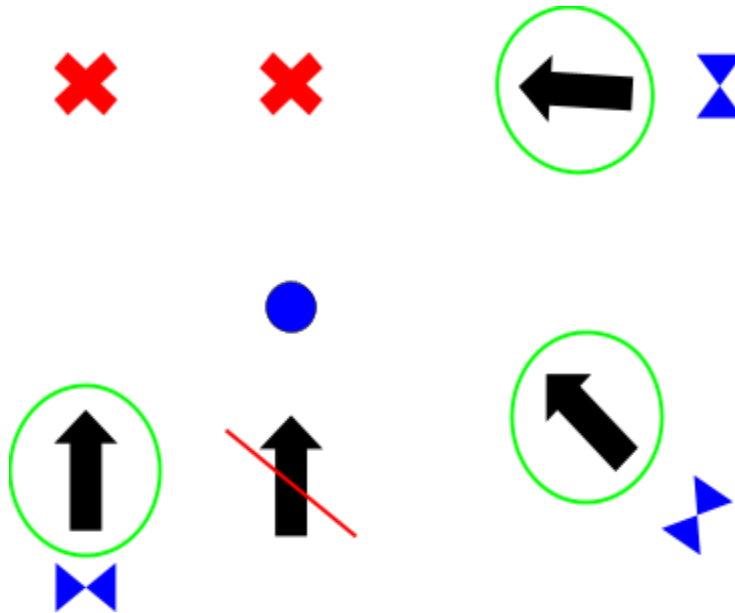
Guns/Cannons	-.50 cal up to 30mm+
Hydra 70	-70mm unguided rockets of various types (HEDP, HEAT, WP)
GBU-12	-500lb laser guided bomb (most common)
GBU-16	-1000lb laser guided bomb (good for larger areas or groups of vics)
GBU-10	-2000lb laser guided bomb (entire compounds villages will disappear)
EGBU	-GBU's with added GPS guidance capability
JDAM	-GPS guided bombs, various sizes and types
AGM-65	-Air to Ground missile, various types (laser or IR guidance)
AGM-114	-Air to Ground missile, various types (laser or IR/radar guidance)
AIM-9	-Air to Air Sidewinder missile, various types, AIM-9X most common
Rockeye	-Unguided cluster bomb
CBU	-Sensor fused anti-tank cluster bomb, various capabilities

Building a CAS Friendly AO

The CAS Controller's task starts the moment they go live in the server. You need to understand the ground commander's intent, planned objectives, axis' of advance, RV's, OP's, locations of enemies and reinforcements, etc. This all contributes to building up the AO for CAS. Specifically via the map. Once objectives, movements, and friendly positions are planned, **IPs**, **BPs**, **CPs**, and **HAs** can be established.

-**CPs** and **HAs** are solely based on convenience in and out of the AO, and relative safety of the aircraft. Things to consider: Terrain, emergency landings, urgent CAS, IDF support and their Gun Target Line (if any), relative position from planned **IPs** and **BPs**.

-**IPs** require analysis of anticipated friendly positions and routes, terrain, building compositions, and threats to the aircraft. Attack headings should typically be offset from friendlies by distance, or by angle. Headings directly over or toward friendlies should be avoided at all costs, as ordinance is more likely to be dropped long or short than it is left or right.



-**BPs** are selected with similar considerations to **IPs**, and are best situated behind some form of cover or concealment. Terrain, buildings, trees, are all acceptable. This allows the rotary gunship to 'pop-up' from cover, and engage the target, before going back down into cover. Hellfire missiles can also be employed using Lock On After Launch (LOAL- LOW & HI) capabilities. After launch, missiles will gain 90m/300m (LOW/HI) to avoid terrain obstacles, lock on to the target after gaining visual contact, and strike all while the aircraft is hidden behind cover. Incredibly powerful tool at the hands of a capable aircrew.

Multiple **IPs** and **BPs** are recommended to ensure CAS solutions are prepared for all possibilities. 3-4 **IPs**, and 2-3 **BPs** are usually sufficient.

At this stage, it's very important to understand both the skill level and realism interest level of your pilots. Good, high fidelity pilots will enjoy using the full extent of what you may prepare. New pilots may be unable to, and casual pilots may not enjoy it. Know your audience and plan accordingly. The author has been handed IRL ple planned fire packages to use in Arma missions, and used well labelled CAS maps prepared ahead of the op for the sake of enjoyment and fidelity.s. Masochism at time others prefer to drop a dot on the map. You can still use all the tools and procedures discussed here, but those tools only go so far as the other party on the radio with you is willing and able. Give a grid ref, read the 9 line, and if the pilot wills it, put a dot on the map and watch it burn.

CAS Check-in

Once an aviation asset is airborne and established at the HA/CP, or in a safe area on standby, the CAS check-in can be used to establish comms with you (the controller).

Format of a check in is fairly loose, but the pilot will inform you WHAT they are flying, and what ordinance they have onboard. Sensors, available flight time, etc may also be relayed.

Buckshot: "Vanguard, Buckshot, holding at CP Alpha. Advise when ready for **CAS check-in, OVER**" -*That is a warno. It provides context for the upcoming comms.*

Vanguard: "Vanguard, send **check-in, OVER**"

Buckshot: "Buckshot is in an A-10 Charlie, equipped with 4x Maverick, 12x Paveway, 19x Hydra, and guns. BREAK." "Time on station is 40 minutes, OVER."

Vanguard: "Vanguard, copy, A-10 Charlie with 4x Maverick, 12x Paveway, 19x Hydra, and guns, time on station 40 minutes. BREAK" "bla bla bla provides tasking or situational update"

Area of Operations Situational Awareness (SA) Update

Once the pilot has completed their check-in, typically you want to provide them with an SA update. Info to include in the SA is location and vector of friendlies, possible threats to the aircraft (AA, AAA), composition of enemies, and what their taskings will likely be (this is called a game plan). All of this info gets the pilot oriented in the battle space, prepared for any defenses if needed, and already thinking about the weapons they will employ.

Altitude brackets, flight restrictions, and other information is also passed along in an SA update. As always with comms, get in touch first, and then send the info in the next transmission.

Vanguard: "Buckshot, Vanguard, advise when ready for **SA** update, OVER"

Buckshot: "Buckshot send traffic, OVER."

Vanguard: "Vanguard, bla bla bla, OVER."

Buckshot: "Buckshot, *reads back exactly what was said by Vanguard*, OVER."

Vanguard: "Vanguard, Good readback, OUT."

CAS Call

Identifying taskings that you can bring CAS in on is a balancing act. This is where the aspect of curating occurs for the Controller. If ROEs, Commander's desire, CAS capabilities, and target type all align, then you will be in a position to 'Spin Up' a CAS call. Information to obtain is found within the 9-line.

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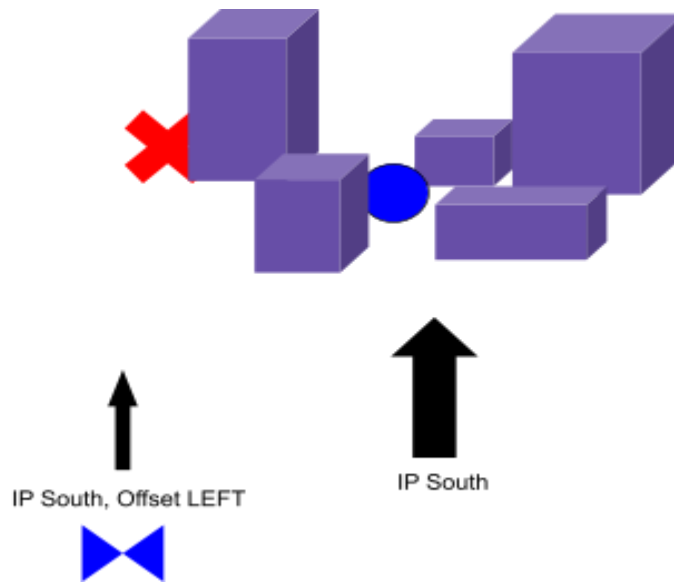
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- The offset is defined as LEFT or RIGHT, with a distance relative to the direction and position of the IP.
- Target description is very self explanatory. Context of the target is also helpful, “in the open”, “third story”, and somewhat rolls into Grid / Location.
- Grid references are often read backwards, or as the intersection of the grid lines. Read along the bottom, and then up the side. In the door, up the stairs. Eg 060, 051 refers to a 100m x 100m area on the map, not a single ‘point’. Location can also be relative to another known object or position. “50 metres WEST of last engagement/target” can be a very good reference for the pilot.
- Ordinance and mark are based on context and capabilities. In the above example, a Maverick is best because of reduced collateral damage. Friendly position doesn’t allow them to place any smoke or lasers, so “no mark, own sensors” would be used. Both of these are entirely situational. Right tool for the right job and as long as it delivers good effect, it works.
- Egress is also situational. Try to direct pilots away from additional enemies or threats after they release ordinance. Can be “at your discretion” if direction is not critical.

Once you have the 9-line made up, you can initiate the CAS call.

The warno for a CAS call is fairly simple, just inform the pilot you have a new tasking for them and request they advise when ready to copy the 9-line. As part of this, you can advise what the game plan is for the tasking (bomb on target, bomb on coordinate, gun run, rocket run, and what IP/BP you will be using so they can begin to position themselves).

Sending The 9-line

1. **IP**
2. **Heading / Offset**
3. **Distance from IP**
4. **Target Elevation**
5. **Target description**
6. **Grid / Location**
7. **Mark on Target**
8. **Friendlylies & Mark**
9. **Egress**

Pilots MUST provide readback of these 9 lines before continuing to remarks.

Remarks after good readback are optional, though helpful. Nearest friendly's location, possible threats to the aircraft, altitude bracket they are allowed to be in, danger close to friendlylies?

Altitude brackets are used to control multiple types of CAS aircraft, avoidance of AA/AAA, deconfliction with IDF, etc. 0-100m for rotary, and 500m and above for fixed wing are common.

The CAS Run

Once the 9-line has been passed, readback complete, and any remarks acknowledged, the Controller should advise the pilot they are “cleared to push” or just told to advise when inbound.

Ground commanders want to know when CAS is inbound, and you will want to know when the pilot passes through the IP to determine when to relay that information to ground forces.

This is best done when passing through the IP mark before the pilot has visual contact. You can then call “continue”, which is authorization for the pilot to acquire visual contact and weapons lock on, NOT WEAPONS RELEASE.

Once they have visual contact, lock-on, etc. the pilot can call “Contact”, or “locked on”, and provided the target area is still valid, and you are certain they’re lined up with the right target, you can then “Clear hot” if you are authorized to do so.

This is all fairly optional depending on the setting. The following is not.

Once cleared hot, the pilot will call “Pickle”, “Rifle”, or “Guns”, so you know the ordinance has been dropped. If this is not called, it is assumed it was a ‘cold’ run, and nothing was dropped.

This is critical to relay to friendlies when to get down and keep in cover, but allow them to keep shooting as long as possible.

Once the impact has been observed, the controller can call “shack”, or “good effects”, or any other remarks about the drop (short, long, etc).

Buckshot: “Vanguard, Buckshot, inbound through IP Alpha.”

Vanguard: “Vanguard, continue.”

Buckshot: “Buckshot, contact.”

Vanguard: “Vanguard, cleared hot.”

Buckshot: “Buckshot, Pickle, egressing South.”

Vanguard: “Vanguard, copy pickle.”

It is critical that the comms rapport not be interrupted by other callsigns/assets during the terminal phase of control. Callsigns and ‘OVER’ can be dropped during established rapport such as this, provided the comms net is clear of other traffic as it should be.

Battle Damage Assessment

Once the impact and effect has been observed, the BDA can be provided by either party.

“Target destroyed”, “50%”, “10 for 10” are all examples of a BDA. The goal is to relay how effective the ordinance drop was on the intended target.

Depending on the BDA, the controller may request immediate re-attack. The re-attack can be the exact same, or specify a change in weapons type, direction of attack, or an adjustment to the target. Examples of changes can be “from the reciprocal” (opposite direction), “switch to guns”, or “shift fire 50m West of previous attack”.

Types of Control(1,2,3)

The US Airforce and DoD describes these types of controls and when to use them best. We aren't reinventing the wheel with this aspect of CAS control, so here it is:

"There are three types of CAS Terminal Attack Control: Types 1, 2 and 3. JTACs use Type 1 control when the situation requires them to visually acquire **both** the attacking aircraft and the target under attack. Type 2 control will be used when the JTAC maintains control of individual attacks, but **either*** visual acquisition of the attacking aircraft or the target at weapons release is not possible or required. Type 3 control may be used when the tactical situation indicates that CAS attacks impose a low risk of fratricide. When commanders authorize Type 3 control, JTACs grant a "blanket" weapons release clearance to an aircraft or flight attacking a target or targets which meet the prescribed restrictions set by the JTAC. The JTACs will broadcast the type of control in use upon aircraft check-in. Type 1 is the default method of control because it is the most restrictive and therefore the least likely to result in incorrect targeting."

* [one or the other, not neither]

Bolding has been used to emphasize certain key aspects of the differences. Type of control can be relayed prior to the 5-line.

Further notes on Type 3:

"The JTAC monitors radio transmissions and other available digital information to maintain control of the engagement. The JTAC maintains abort authority. Observers may be used to provide targeting data and the target mark during Type 3 control. Type 3 is a CAS TAC procedure and should not be confused with terminal guidance operations or AI. Missions attacking targets not in close proximity to friendly forces, and beyond the range requiring detailed integration with surface fires and maneuver, should be conducted using air interdiction (AI) procedures vice CAS."

Very good write up of all three types of control:

https://www.doctrine.af.mil/Portals/61/documents/Annex_3-03/3-03-D19-LAND-Type-Terminal-Ctrl.pdf

One last thing, the official US Military JFIRE is the irrefutable BIBLE when it comes to CAS and IDF control. This is now a somewhat outdated publication (2007, newest version is 2019), however most of the information is still relevant and in use:

<https://info.publicintelligence.net/MTTP-JFIRE.pdf>